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When Costco Comes to Town, Can Adjusting Product Variety Shield Incumbent Stores From Sales Loss? Insights From Retail Collocation, Category, and Preexisting Competition

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Abstract

When a competitor enters a market, incumbent retail stores often adjust their product variety to avoid sales loss. However, no empirical evidence has been provided in the literature to justify the value of such product variety adjustments after a rival's entry. One difficulty is that the change in consumer demand due to rival's entry and changes in product variety simultaneously affect incumbents' sales. When changes in sales at incumbent stores are observed after rivals' entries, it is unclear whether the sales changes result from product variety adjustments or changes in consumer demand directly resulting from the entry. Even if adjusting product variety can potentially mitigate sales loss due to rivals' entries, it is unclear whether such product variety adjustments provide a universal solution to all incumbent stores. This paper applies the difference-in-difference research design to develop a system of equations for product variety and sales at incumbent stores after rivals' entries by exploiting the entries of warehouse clubs. The findings reveal that, on average, product variety at incumbent stores increased after the entries. This increased product variety mitigates the sales decreases caused by the entries with a diminishing return. After the entry of a warehouse club store, incumbent stores collocated with the new store have less sales loss than the noncollocated stores due to the trade-off between the substitution and agglomeration effects. Product variety adjustments at collocated incumbent stores experience greater sales benefits than noncollocated stores. Moreover, the sales loss that was mitigated by increased product variety at incumbent stores within the retail category of entrants is larger than others after the entry of a warehouse club store. Interestingly, the mitigation effect is not significant for certain incumbents, such as convenience and drug stores, which belong to different retail categories from entrants. This study also explores the heterogeneous effects by investigating preexisting competition and different warehouse club chains. The empirical findings offer valuable insights to managers of incumbent stores on how to adjust product variety to reduce sales loss resulting from a rival's entry.

Keywords

Product Variety, Collocation, Agglomeration, Retailing, Incumbent Responses, Market Entry, Sales Loss

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1 Introduction

New competitors to a market often disturb the existing market equilibrium, intensify competition, and often reduce the sales of incumbents. In the retail industry, price and nonprice strategies have been commonly applied by incumbents. Product variety has been recognized in the literature as a popular nonprice strategy in response to changes in competition (Ren et al., 2011, 2019; Watson, 2009). Whether and how they should adjust product variety depends on the contribution of product variety adjustments after entry to their sales. One difficulty in empirically quantifying the value of such product variety adjustments is that both the change in consumer demand due to the rival's entry and the incumbent retailers' product variety decisions simultaneously affect incumbents'

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sales. When changes in sales at incumbent stores are observed, the sales changes result from mixed effects of product variety adjustments and changes in consumer demand directly from the entry.

This paper contributes to the literature on product variety management and market competition (Fisher and Ittner, 1999; Ramdas, 2003; Sanders and Wan, 2018; Zhou and Wan, 2017a). The empirical analyses disentangle the mixed effects of product variety adjustments and rivals' entries on sales, then quantify the value of product variety adjustments after rivals' entries by exploiting entries of warehouse club stores (such as Costco, Sam's Club, and BJ's). Intuitively, sales at incumbent stores competing with entrants are expected to decrease after the entries because entrants generate a substitution effect on incumbent stores, which attracts consumers away from incumbents and reduces their sales (Chang, 1996; Seamans, 2012, 2013). When incumbent stores increase their product variety after rivals' entries, a smaller reduction in sales indicates the effectiveness of this product variety decision. To capture these relationships, I developed a system of equations (one for product variety, the other for sales) using the difference-in-difference (DID) approach to answer the first set of research questions: How do incumbent stores adjust product variety after rivals' entries? How much do product variety adjustments alleviate the negative impact of rivals' entries on sales at incumbent stores? And does this moderating effect follow a linear or nonlinear pattern?

A rival's entry leads to different influences on collocated and distant incumbents (Ren et al., 2011, 2019). When a new warehouse club store is opened, it intensifies competition in the market and generates a substitution effect on incumbent stores, which attracts consumers away from incumbents and reduces their sales (Chang, 1996; Seamans, 2012, 2013). Meanwhile, based on the literature on agglomeration (Alcácer, 2006; Alcácer and Chung, 2007, Rawley and Seamans, 2020), sales at the incumbent stores collocated with the newly opened store (hereinafter referred to as collocated stores) may also benefit from the agglomeration effect, which attracts more demand to the store location and reduces the shopping costs of consumers who shop at stores in close proximity (Dellaert et al., 1998, Rosenthal and Strange, 2003). Increased consumer traffic raises sales at retail stores (Perdikaki et al., 2012). Thus, although all incumbent stores in the market face a substitution effect of rivals' entries that lowers sales, the change in sales at the collocated stores after rivals' entries depends on the trade-off between the substitution and agglomeration effects. Accordingly, the mitigation of product variety on incumbents' sales loss after rivals' entries may vary across collocated and noncollocated stores.

Additionally, in an imperfectly competitive market, firms can be clustered into different categories based on their heterogeneity (Cattani et al., 2017; Porter, 1979, 2008). The literature on categories and competition suggests that firms' decisions and their influences vary depending on their categories (Suarez et al., 2015). The value of product variety adjustments for incumbent stores after competitors' entries may also depend on the retail categories to which they belong. Newly entered warehouse club stores are in the same category as some incumbents but are in different categories from other incumbents. In the retailing industry, significant differences exist among various retail formats (Bell and Lattin, 1998; Hansen and Singh, 2009), such as warehouse clubs, supermarkets, mass merchandisers, convenience stores, and drug stores. Retail formats within the same category as the new warehouse clubs experience the impact of entry and benefit from product variety adjustments differently from other retail formats in cross-categorical competition with the new warehouse clubs. By exploring the collocation and within-categorical competition between incumbent stores and entrants, this study addresses the second set of empirical questions: After competitors enter the market, should all incumbent stores apply a unified product variety adjustment decision? If not, how should product variety adjustments be tailored to various incumbents after rivals' entries?

Furthermore, this paper explores factors that could affect the influence of product variety adjustment after competitors' entries, such as the preexisting competition and different chains of newly entered warehouse clubs. The impact of entry on incumbents' sales and the mitigation role of product variety adjustments are likely to be heterogeneous across these factors. By analyzing these factors, this empirical study addresses the question: How should incumbent stores adjust their product variety after entry, taking into account their features in the market, such as preexisting competition and different warehouse club chains?

The data were collected from a beverage production company, warehouse club chains' websites and annual reports, the U.S. Census Bureau, and the U.S. National Oceanic and Atmospheric Administration (NOAA). The empirical results show that product variety increased after entry and, on average, sales at the incumbent stores decreased by 17.8% after the entry of competitors. However, after increasing the product variety by one unit, the sales loss at stores was reduced by 0.2%. This means that compared to an average 17.8% sales decrease after entry, sales at an incumbent store, which adds 10 more nonoverlapping stock-keeping units (SKUs), will have a sales decrease of 15.8% (calculated by $17.8\% - 0.2\% \times 10$). Relatively, the decrease in sales at this incumbent store is 88.8% (calculated by 15.8%/17.8%) of the sales decrease at stores with no product variety adjustment after rivals' entries. In addition, the mitigation effect of product variety adjustments demonstrates a diminishing rate.

Furthermore, decreases in sales after rivals' entries at collocated stores are found to be less than those at noncollocated stores. This finding provides empirical evidence for the agglomeration effect associated with rivals' entries. The mitigation effect of increased product variety on sales loss due to rivals' entries at collocated stores is larger than that at noncollocated stores. Incumbent stores in within-categorical competition with entrants experience a stronger effect of product variety adjustments than stores in cross-categorical competition with entrants. More interestingly, increases in product variety mitigate sales decreases due to rivals' entries at within-categorical stores but not at cross-categorical stores. Therefore, simply raising product variety following a rival's entry does not guarantee sales benefits for all incumbents.

Moreover, the analyses explore the mitigation effect of product variety across external features in the market. A higher level of preexisting competition weakens the mitigation effect of product variety on the incumbents' sales loss after entry. And increases in product variety alleviate the sale reduction at incumbent stores due to entries of Costco stores more than entries of BJ's stores.

Propensity score matching (PSM) is applied to match stores to reduce the potential selection bias of retail stores in the sample. The three-stage least squares with the instrumental variable method is used to address the potential endogeneity associated with product variety decisions. In addition, tests are conducted for the parallel trend assumption and placebo tests in the DID framework. Additional analyses rule out the alternative explanation that changes in product variety and sales at incumbent stores result from the entry/exit of retail stores other than warehouse club stores.

This study complements the growing literature on product variety and competitive dynamics by evaluating the effect of product variety adjustments on incumbents' sales after competitors' entries. The paper also investigates the nonlinearities of the effects of product variety, heterogeneous effects for store locations, within- and cross-categorical competition, preexisting competition, and entries of different warehouse club chains. The findings provide practical guidance for managers of incumbent retail stores in adjusting their product variety after a rival's market entry. The contributions are further explained in the last section.

The structure of the paper is designed to provide a clear and logical flow from theoretical foundations to practical implications. Section 2 presents the literature review. Section 3 focuses on theory and hypothesis development, where the conceptual framework is built and hypotheses are formulated. In Section 4, the data and measures are explained, detailing the data sources and variables. Section 5 introduces the empirical models, explaining the methodological approach used to test the hypotheses. The estimated results are presented in Section 6, followed by Section 7, which provides robustness checks to ensure the reliability of the findings. Section 8 examines heterogeneous effects across competition categories and entrant chains. Section 9 outlines the managerial implications, highlighting practical insights for practitioners. Finally, Section 10 offers a conclusion, contributions to the field, and limitations of the study.

2 Literature Review

In the operations management literature, product variety management is important to production (Fisher and Ittner, 1999; MacDuffie et al., 1996; Ramdas et al., 2003), inventory (Fisher et al., 1999; Moreno and Terwiesch, 2017; Ramdas, 2003; Wan and Sanders, 2017), competition (Alptekinoğlu and Corbett, 2008; Wan and Li, 2024), sales (Lancaster, 1990; Li and Wan, 2023; Ton and Raman, 2010; Wan et al., 2014), and overall profit (Caro and Gallien, 2007; Wan and Dresner, 2015). Meanwhile, the strategic management literature has identified product variety as a common strategy in response to competition (Dixit and Stiglitz, 1977) and raising the entry barriers by filling potential market gaps (Bayus and Putsis Jr., 1999).

Product variety, in general, helps firms increase sales by satisfying heterogeneous consumer preferences, raising consumer welfare, and even stimulating variety-seeking behavior (Bernstein and Guo, 2023; Brynjolfsson et al., 2003; Kekre and Srinivasan, 1990; Lancaster, 1990, Tayur, 2017). However, in the literature on the impact of product variety on sales, the relationships between product variety and sales have commonly been studied in stable markets with no entrants (Brynjolfsson et al., 2003; Ramdas, 2003; Wan et al., 2012). The impact of rivals' entries on sales could be so dominating that the influence of adjusting product variety becomes too trivial to mitigate the sales loss due to entry. Few studies provide empirical evidence for the value of product variety in alleviating sales loss after entry. This paper quantifies the effect size and identifies the nonlinear pattern of product variety adjustments in mitigating the negative impact of market entry on incumbents' sales. Furthermore, this study extends the prior literature by investigating how product variety adjustments contribute to sales loss due to rivals' entries at stores collocated with or in within-categorical competition with entrants differently from other stores.

3 Hypothesis Development

3.1 The Base Relationship: Effects of Rivals' Entry on Incumbents' Sales

Sales value is the product of sales quantity and price, both of which are affected by new entrants to the market. Regarding sales quantity, the entry of a warehouse club store introduces a new shopping site to consumers and generates a substitution effect on the sales quantity at incumbent retail stores. With the substitution effect, incumbent stores are expected to experience a reduction in sales quantity.

The product prices at incumbents may decrease, remain unchanged, or even increase after a rival's entry. Due to increased competition resulting from entrants, the incumbents may lower prices in response to entry (Seamans, 2013; Simon, 2005). A limit pricing strategy sends a low-cost signal to entrants and helps retain customers. With no entrants, a lower price may increase incumbents' sales quantity by attracting more customers. After a rival's market entry, a lower price may mitigate the decreases in incumbents' sales quantity due to the entry. Meanwhile, some retailers may avoid price competition and maintain their preexisting pricing. For example, Best Buy and Circuit City stores did not significantly change their prices for digital camera and TV products when competition varies in the market (Ren et al., 2011, 2019). Since sales quantity at incumbents decreases after a rival's entry due to the substitution effect, sales (as the product of quantity and price) would decrease when product price decreases or remains unchanged after entry.

Incumbent stores may also increase prices if their products are differentiated from the products offered by competitors. In our context of nonalcoholic beverages, however, the products at competing incumbent stores are not substantially different. After a rival's entry, if an incumbent increases prices of products similar to those at other stores, demand will switch not only to entrants but also to other incumbent stores. As a result, an increase in product price at incumbents would magnify the sales loss after entry. Hence, considering the effects of rivals' entries on both sales quantity and price, the entry lowers sales at incumbents. The effect of entry on incumbents' sales is proposed to serve as the base relationship for the hypotheses in the following subsections.

Hypothesis 1. After a warehouse club store enters a market, sales at the incumbent stores decrease.

3.2 The Mitigation Role of Product Variety Adjustments in the Rival Entry–Sales Relationship

For retail chains, headquarters and store managers make product variety decisions jointly. For example, a Wal-Mart store in one location may carry different breakfast cereal products than its stores in other locations. At retail stores, products are introduced in retail flyers to consumers. New products are often marked in the price tags.

The motivation for reducing product variety is to reduce overlapping products with entrants (Ren et al., 2011). When facing competition generated by the warehouse club's entry, incumbents are likely to increase nonoverlapping product variety by carrying unique products. Increases in nonoverlapping product variety affect both sales quantity and product price at incumbent stores. Regarding sales quantity, when increasing product variety, incumbents differentiate from entrants and weaken the substitution effect of entry on sales quantity at incumbents by attracting consumers from competing stores and reducing the number of consumers who switch to the new warehouse club. Regarding product price, increased nonoverlapping product variety may enable stores to charge relatively higher prices for their products (Amaldoss and He, 2010; Lancaster, 1990), because the differentiated products lessen the direct price competition between stores. Although incumbents do not necessarily increase their product price after entry, the potential price-raising effect of increased product variety could soften their price reduction decision in response to entry. Therefore, increased product variety at incumbent stores mitigates decreases in their sales quantity and price after a rival's entry and, in turn, alleviates their sales loss.

Hypothesis 2a. After a warehouse club store enters a market, sales at the incumbent stores that increase nonoverlapping product variety decrease less than at other incumbent stores.

The mitigation effect (a smaller decrease in sales) of the increased nonoverlapping product variety at incumbent stores may not be linear and presents a diminishing return. As product variety increases, the cannibalization between products increases (Bayus and Putsis Jr., 1999; Kekre and Srinivasan, 1990). Adding new products to a product category at retail stores cannibalizes sales of existing products. Moreover, the marginal consumer utility raised by additional products decreases with a higher product variety because too much variety may lead to selection confusion (variety fatigue) (Thompson et al., 2005). Thus, the product cannibalization and variety fatigue associated with higher product variety reduce the marginal mitigation impact of increased product variety on the sales loss after entry.

Hypothesis 2b. After a warehouse club store enters a market, the sales benefit (the smaller decrease in sales) of increasing nonoverlapping product variety at the incumbent stores has a diminishing return.

3.3 Collocated versus Noncollocated Stores: Substitution and Agglomeration Effects

In addition to the substitution effect that lowers the sales at incumbent stores (discussed in the previous section), a warehouse club's entry also generates an agglomeration effect on the incumbent stores that are collocated with it (Alcácer, 2006; Alcácer and Chung, 2007; Rawley and Seamans, 2020). When a new warehouse club store attracts consumer traffic to its locations, the additional demand may benefit the incumbents collocated with the entrant (Ren et al., 2011). For example, a new Costco store is expected to attract hundreds of consumers a day because they are willing to travel long distances to shop at discounted prices (Alfs, 2012). Consumers often combine shopping purposes and destinations to reduce their shopping time and increase their shopping efficiency (Dellaert et al., 1998). When consumers arrive at the warehouse club's location, they can shop at collocated stores with little added transportation cost. Larger consumer traffic increases sales at these stores (Perdikaki et al., 2012). Incumbent retailers can benefit from heightened demand caused by agglomeration (Fischer and Harrington Jr, 1996).

Since a warehouse club store carries a limited selection of products, shoppers are likely to visit nearby stores for other products after shopping at a warehouse club store. This purchase at the nearby store would not have occurred if a shopper had not traveled to the warehouse club store. Thus, the agglomeration effect associated with the warehouse club's entries attracts new consumers to collocated incumbent stores and increases their sales. While the collocated incumbent stores are influenced by both agglomeration and substitution effects, the noncollocated incumbent stores are affected only by the substitution effect from the entry of a warehouse club. Thus, the collocated incumbent stores are expected to experience fewer sales decreases after rivals' entries than noncollocated stores.

Hypothesis 3a. After a warehouse club store enters a market, sales at incumbent stores collocated with the new store decrease less than sales at other incumbent stores.

Sales at collocated stores may benefit from increased product variety more than at noncollocated stores, because the increased product variety at collocated stores mitigates the substitution effect and enhances the agglomeration effect of the warehouse club's entry. By increasing product variety, collocated and noncollocated incumbent stores all mitigate the substitution effect by preventing consumers from switching to the new warehouse club store. Additionally, increases in product variety enable collocated stores to enhance the agglomeration effect by attracting consumers. A higher product variety in a store cluster of incumbent stores and the newly entered warehouse club store (such as a shopping plaza) attracts more consumers to travel to and shop at their location. Hence, increasing product variety is expected to have a larger sales benefit for collocated stores than for noncollocated incumbent stores.

Hypothesis 3b. After a warehouse club store enters a market, the sales benefits of increasing nonoverlapping product variety at incumbent stores collocated with the new store are larger than those at other incumbent stores.

3.4 Within-Categorical and Cross-Categorical Competition

Although new warehouse club stores potentially raise the competition level in the market, a new store does not necessarily raise competition uniformly to all incumbents. Competition between firms is commonly based on similarities and differences in their resources and consumer pools (Porac and Thomas, 1994). Different retail formats are heterogeneous in their average product variety, price, distance to residential areas, etc. (Bell and Lattin, 1998; Hansen and Singh, 2009). Besides warehouse clubs, consumer packaged goods are commonly sold at supermarkets (such as Kroger and Safeway), mass merchandisers (such as Wal-Mart and Target), convenience stores (such as 7-Eleven and Circle K), and drug stores (such as CVS and Walgreen).

Warehouse club stores have product prices, transportation costs, distance to residential areas, and shopping areas similar to supermarkets and mass merchandise stores. However, convenience and drug stores are different from warehouse club stores in most of these features. Due to their similarities, incumbent warehouse clubs, supermarkets, and mass merchandisers compete with newly entered warehouse club stores more strongly than convenience and drug stores. Sales at incumbent stores in within-categorical competition with new entrants (hereinafter referred to as within-categorical stores) are expected to decrease after entry more than sales at other incumbent stores in cross-categorical competition with entrants.

Hypothesis 4a. After a warehouse club store enters a market, sales at incumbent stores in within-categorical competition with the new store decrease more than sales at other incumbent stores.

Similarly, adjusting product variety after entry does not necessarily generate consistent sales benefits across categories. Increasing product variety helps within-categorical stores differentiate themselves from the new entrants, especially when the newly entered warehouse club stores do not carry a great variety of products. Instead of high product variety, cross-categorical stores (such as convenience and drug stores) offer shopping conveniences, including lower transportation costs, easier product searching, and shorter checkout times (Bell et al., 1998). Convenience and drug stores cannot easily differentiate themselves from the entrants by offering more products. Thus, the following hypothesis is proposed.

Hypothesis 4b. After a warehouse club store enters a market, the sales benefits of increasing nonoverlapping product variety at incumbent stores in within-categorical competition with the new store are larger than those at other incumbent stores.

3.5 Preexisting Competition Before Entry

Although the entry of warehouse club stores generates a substitution effect on incumbents, the preexisting competition faced by incumbent stores varies and moderates the effect of market entry (Newbery, 1998). The competition raised by an entrant in a highly competitive market is not as significant as in a noncompetitive market. The substitution effect generated by an entrant diminishes with the increases in the existing competition in a market. For example, a new Costco store would have more impact on an incumbent store in a market of 10 existing stores than on a store in a market of 1,000 existing stores. Additionally, incumbent stores in a highly competitive market are generally more prepared for competition than those in a noncompetitive market. When a new warehouse club is opened, incumbent stores in a highly competitive market will be proportionally less affected than incumbent stores in a noncompetitive market. Thus, preexisting competition weakens the negative impact of entry on sales at incumbent stores.

Hypothesis 5a. After a warehouse club store enters a market, sales at the incumbent stores in a market with more preexisting competitors decrease less than those at incumbent stores with fewer preexisting competitors.

By providing more nonoverlapping products, incumbent stores can differentiate themselves from the entrants and attract consumers by better fitting their diversified preferences (Berger et al., 2007; Lancaster, 1990). However, the sales benefits of increased product variety at incumbent stores are smaller in a more competitive market before entry. On the one hand, product variety at incumbent stores is possibly greater in a more competitive market. The benefits of an increase in product variety diminish with a higher preexisting product variety level. On the other hand, in a competitive market, consumers are already exposed to various competing strategies, such as advertising, promotion, pricing, etc. Consumers in such a competitive market would be less sensitive to increases in product variety at incumbent stores after entry. Hence, the following hypothesis is proposed.

Hypothesis 5b. After a warehouse club store enters a market, the sales benefits of increasing nonoverlapping product variety at incumbent stores with more preexisting competitors are less than those at incumbent stores with fewer preexisting competitors.

4 Data and Measures

Data for this study were collected from several sources, including (1) proprietary data of product variety, sales quantity, price, and stores' street addresses from the retail customers of a production company of nonalcoholic beverages; (2) open dates of warehouse club new stores from the proprietary dataset combined with information on warehouse club chains' websites and annual reports; (3) demographic data from the U.S. Census Bureau; and (4) weather data from the U.S. NOAA.

4.1 Proprietary Product Variety and Sales Data

4.1.1 Data Sample. Through a collaboration with a U.S.-based production company, product variety and sales data were obtained. This company produces branded nonalcoholic beverages and sells them to retail stores. Although there is an increasing trend of online sales in many industries (Brynjolfsson et al., 2003; Hamilton and Richards, 2009), the firm's product sales through online retailers are relatively small compared to sales through brick-and-mortar retail stores during the time window in the data. The company provided sales records for nine continuous months between June 2010 and February 2011 at all retail stores that sell its products in Washington D.C. and three U.S. states: Maryland, Virginia, and Georgia. The data sample includes 10,770 retail stores that operated throughout the data time window.

4.1.2 Product Variety and Sales. Product variety (PV_{it}) is measured by the number of nonoverlapping SKUs at retail store i in month t, which do not overlap with the products at the newly entered warehouse club stores. The number of SKUs is commonly used in the literature on product variety (Fisher and Ittner, 1999; Ren et al., 2011, 2019; Zhou and Wan, 2017b). Some beverage products in warehouse club stores have different SKUs from those in other retail stores because they are in different pack sizes or containers. To identify nonoverlapping products, we use a unique product indicator in our dataset, which indicates the combination of product trademark, brand, and flavor. By using this product indicator, we filtered the SKUs in the incumbent stores that do not overlap with products sold in newly entered warehouse club stores. Sales ($Sales_{it}$) is the sales value (the product of sales quantity and price) in dollars at retail store i in month t.

4.2 Entry of Warehouse Clubs

Warehouse clubs invest in new stores even after anticipating the existing competition and incumbents' responses. These responses might reduce but not eliminate the benefits of entry. The entries of three warehouse club chains (Costco, Sam's Club, and BJ's) were identified through multiple sources. First, in the proprietary sales data, the first-added months of all retail stores were obtained, which indicate the first month when a retail store was added to the production firm's information system as a business customer. Second, the addresses of these warehouse club stores in the proprietary data were used to locate them on the store locator webpage of these three warehouse club chains (costco.com/warehouse-locations, samsclub.com/locator, and bjs.com/clubLocator). These stores were then contacted to verify their open months. Lastly, these opening months and states of these warehouse club stores were verified in the annual reports of the warehouse club firms. After the above three steps, four warehouse club stores (two Costco stores and two BJ's stores) were identified as newly opened in the three states and Washington D.C. during the 9 months included in the sales data. All of these stores opened in October 2010. Thus, the data sample includes 5 months before the entry (preentry period) and 4 months after the entry (postentry period).

4.3 Geographic Markets and Incumbent Stores

Following prior research in economics and management (Gowrisankaran and Krainer, 2011; Olivares and Cachon, 2009; Ren et al., 2011, 2019; Watson, 2009), the local market was measured using the geographic locations of retail stores. Specifically, the street addresses of the retail stores were used to pinpoint their locations on a U.S. map and a circle was drawn with a Euclidean radius of 10 miles around the focal store to define a local market (Ren et al., 2011). A new warehouse club store, which enters the local market of an incumbent store, potentially competes with it. An example of the local market of an incumbent store is provided in Figure 1. In the data sample, 1,169 retail stores, whose local markets include the entrants, were categorized as the treatment group (incumbent stores that potentially compete with the rival entrants), while the other 9,601 retail stores were categorized as the control group. In the DID framework, the variable $RivalEntry_{it}$ indicates whether an observation of a store is a treated store and occurred after the entry of a warehouse club store. This variable captures the difference between incumbent stores in the treatment group and other incumbent stores in the control group, and the difference between the pre- and postentry time. In addition, the number of retail stores in the market: $ExComp_{it}$ captures the preexisting competition in the market before warehouse clubs' entries.

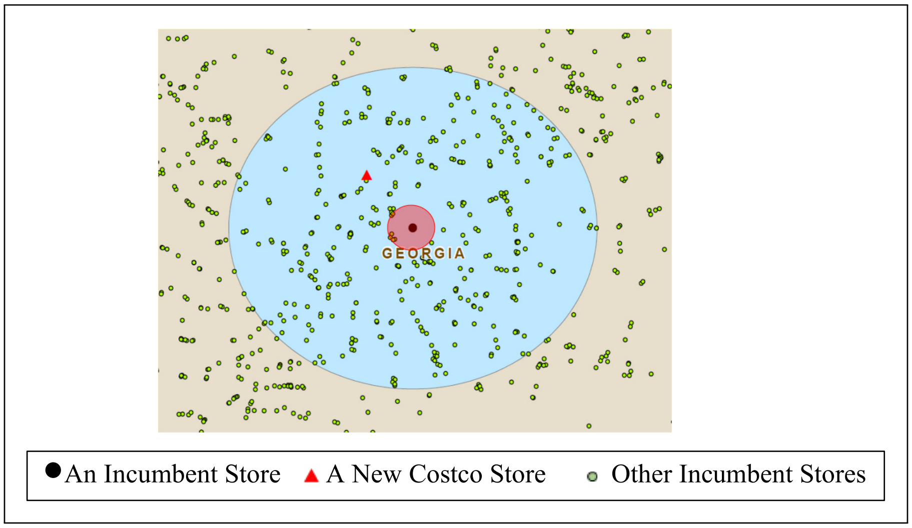

Figure 1. 1.5-mile and 10-mile radii around an incumbent store.

Furthermore, the treated stores are grouped by their location to the entrants. Specifically, a circle with a 1.5-mile radius is drawn around each store. The stores that have the new warehouse club stores within the circles of a 1.5-mile radius are grouped as treated collocated stores. Stores that have the new warehouse club stores within the 10-mile radius circle but outside of the 1.5-mile radius circle are classified into treated noncollocated stores. A dummy variable $Collocate_i$ is coded 1 for collocated stores and 0 otherwise. Robustness tests are conducted with local markets of different radii (such as 15 miles for the local market and 1 mile for collocated stores). The findings remain unchanged.

4.4 Matching and Demographic Data

The differences between stores in the treatment and control groups may affect their product variety responses and sales, which, in turn, lead to selection bias. A three-step matching procedure was applied to reduce potential selection bias. First, since the competition and consumer demand vary across retail formats (Bell and Lattin, 1998; Gonzalez-Benito et al., 2005), the treated stores and control stores are matched in the same retail format, such as warehouse club, supermarket, mass merchandiser, convenience, and drug stores. Second, the matched control stores are constrained to be located in the same designated market area (DMA) with a treated store, so that consumers of stores in both groups are exposed to the same TV advertising and similar demand shifts in the local markets. Third, within the retail formats and DMAs, the PSM is applied to match the treated stores with stores in the control group based on store features, demographic, and economic variables collected from the 2010 U.S. Census.

The three-step matching procedure matched 987 unique treated stores with 987 unique control stores in the data sample. As shown in Table 1, the store features and demographic factors in the matched treatment and control groups are similar, which reduces the potential selection bias in the sample. The *t*-tests suggest the differences between these variables of matched stores in the two groups are not significantly different from zero. Product variety and average sales at stores in the treatment and control groups are plotted in Figure 2. Before the entry, product variety and sales were close between the stores in the two groups. After the entry, product variety increased and sales started to decrease in the treatment group. This plot offers visual evidence for the parallel trend assumption. The trend in product variety and sales was tested in the robustness check section of this study. These plots also provide model-free evidence for changes in sales after entry. Econometrics models were developed under the DID framework in the next section.

The shopping area, price, and distance to the residence area for warehouse clubs, supermarkets, and mass merchandisers are more similar to those associated with new warehouse club stores than those for convenience stores and drug stores. Thus, they are in within-categorical competition with new warehouse club stores and indicated by a dummy variable $WithinCat_i$. The main variables are explained in Table 2 with their descriptive statistics in the matched data sample with 17,766 observations.

The correlations between variables are provided in Table A1 in E-Companion A. As expected, market entry is negatively associated with sales at incumbent stores, and product variety is positively associated with sales. While most variables are correlated with sales, the correlation coefficients between other variables are relatively low. In addition, the variance inflation factors (VIFs) for all explanatory variables in the empirical models are calculated for the consideration of multicollinearity. These relatively small VIFs (less than 4) alleviate multicollinearity concerns.

Table 1. Matched retail stores in treatment and control groups.

	Matched treatment group		Matched control group		*t*-Tests for comparison
Number of stores	987		987		1,974
	Mean	Std. Dev.	Mean	Std. Dev.	*t*-Statistic
Product variety (SKUs)	62.583	39.459	64.008	41.434	0.783
Average price ($)	18.658	2.581	18.590	2.393	0.609
Number of rival stores (100)	2.811	1.935	2.796	1.902	0.174
Population (1,000)	121.824	26.236	123.053	32.826	−0.919
Median household income ($1,000)	74.856	33.759	73.214	34.257	1.073

Note. Std. Dev. = standard deviation; SKU = stock-keeping unit.

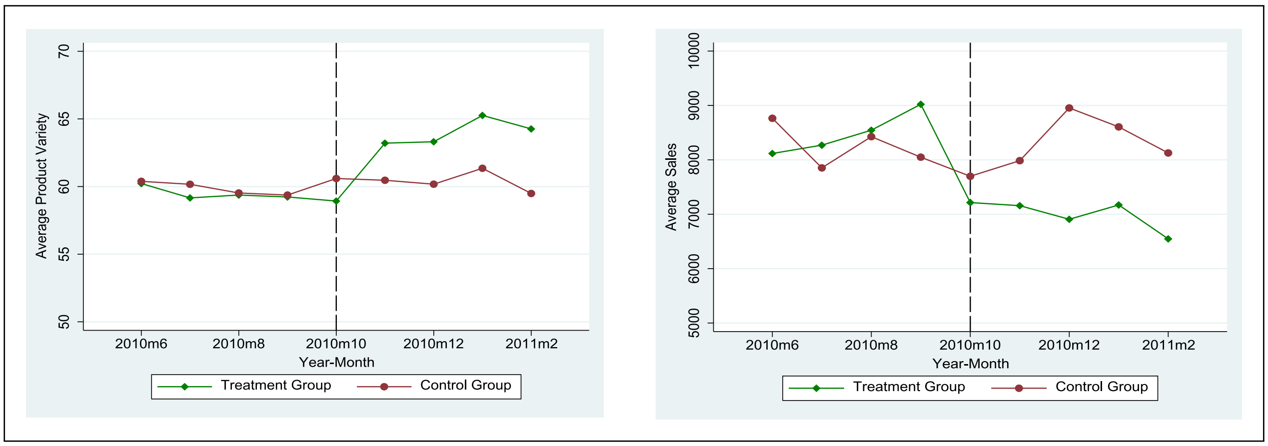

Figure 2. Sales at treated and control stores over time.

Table 2. Descriptive statistics for variables of the matched treated and control stores.

Variables	Description	Unit	Mean	Std. Dev.	Min	Max
$Sales_{it}$	Sales value at retail store *i* in month *t* for products from the focal supplier, which is the product of sales quantity and price.	dollar	7,906.376	13,105.170	235	77,574
$RivalEntry_{it}$	Dummy variable that indicates whether incumbent store *i* has the entrant in its local market in month *t* after the entry.	0/1	0.222	0.416	0	1
PV_{it}	Number of nonoverlapping SKUs at retail store *i* in month *t*.	SKU	53.054	40.701	0	189
$Collocate_i$	Dummy variable that indicates whether store *i* is a collocated incumbent store to the entry.	0/1	0.012	0.107	0	1
$WithinCat_i$	Dummy variable that indicates whether store *i* is in within-categorical competition with the newly entered warehouse club.	0/1	0.281	0.459	0	1
$ExComp_i$	The preexisting competition level (the number of incumbent retail stores) in the local market of store *i* before entry.	100	2.808	1.912	0.107	11.763
$Population_i$	Population in the market of store *i*.	1,000	122.671	34.785	15.988	377.192
$Income_i$	Median household income in the market of store *i*.	$1000	74.349	32.547	21.683	201.615
$Temperature_{it}$	Average temperature in the area of retail store *i* in month *t*.	°C	15.844	9.216	−8.05	30.3

Note. Std. Dev. = standard deviation; SKU = stock-keeping unit.

4.5 Weather Data

Weather conditions are expected to affect product variety decisions (Tian et al., 2018). Daily weather data from the U.S. NOAA were collected. NOAA provides the coordinates of these stations. These weather stations were matched to retail stores by the closest distance. Next, the monthly average values of the temperature ($Temperature_{it}$) were calculated to match the monthly product variety and sales data of retail stores. Temperature is not significantly correlated with sales but significantly and positively correlated with product variety. Thus, $Temperature_{it}$ is included in the model for product variety, not in the model for sales in Section 5.

5 Empirical Models

5.1 Base Model

In this section, I develop a system of two equations: equation (1) was developed for the impact of entry on product variety decisions of incumbent stores, and equation (2) was developed to examine the effect of entry on sales and the mitigation effect of product variety adjustments on the relationship between the entry and incumbents' sales. The DID approach is applied to capture the influence of entry. PV_{it} is the nonoverlapping product variety at incumbent stores for the products from the focal supplier. $Sales_{it}$ is the sales value at incumbent stores for the products from the focal supplier.

$$PV_{it} = \alpha_0 + \alpha_1 RivalEntry_{it} + \alpha_2 Temperature_{it} + store_i + month_t + e_{it} \quad (1)$$

$$\log Sales_{it} = \beta_0 + \beta_1 RivalEntry_{it} + \beta_2 PV_{it} + \beta_3 RivalEntry_{it} \times PV_{it} + store_i + month_t + \varepsilon_{it} \quad (2)$$

where $RivalEntry_{it}$ captures the difference between treated and control incumbent retail stores and the difference between the pre- and postentry periods. The coefficient α_1 in equation (1) captures the effect of entry on product variety. To address the potential endogenous concern on product variety, the temperature ($Temperature_{it}$) is included as the instrumental variable for product variety. The manufacturer included in this study introduced more product variety to areas and in months with high temperatures. Retailers also vary their product portfolio when temperature changes over time. Meanwhile, temperature does not affect sales (its correlation coefficient with sales is 0.009). The strength of this instrumental variable and the exclusion restriction are explained in Section 6. Fixed effects for stores and months ($store_i$ and $month_t$) were included to capture the variation across stores and over time.

In equation (2), natural log-transformation was applied to $Sales_{it}$ for a closer approximation of normal distributions. The equation was developed using a log-linear regression expression between $Sales_{it}$ and $RivalEntry_{it.}$ Thus, the coefficient β_1 indicates the average percentage changes in sales at incumbent retail stores after the entries of warehouse club. When the interaction term $RivalEntry_{it} \times PV_{it}$ is not included in the model, the coefficient β_1 in equation (2) implies the main effect of the entry on sales, which is expected to be negative in hypothesis 1. Similarly, when $RivalEntry_{it} \times PV_{it}$ is not included in the model, the coefficient β_2 suggests the main effect of nonoverlapping product variety on sales, which is expected to be positive. The coefficient for $RivalEntry_{it} \times PV_{it}(\beta_3)$ suggests the mitigation effect of product variety adjustment on the entry–sales relationship. As stated in hypothesis 2a, β_3 is expected to be positive.

5.2 Nonlinear Mitigation Role of Product Variety Adjustments

The nonlinear effect of nonoverlapping product variety is further explored by adding the quadratic term of product variety PV_{it}^2 and its interaction term $RivalEntry_{it} \times PV_{it}^2$ to equation (2) and forming equation (3), if the coefficient β_3 is significantly positive and β_5 is significantly negative, then hypothesis 2b is supported. Equations (1) and (3) were jointly estimated as a system to investigate the nonlinear mitigation effect of product variety.

$$\log Sales_{it} = \beta_0 + \beta_1 RivalEntry_{it} + \beta_2 PV_{it} + \beta_3 RivalEntry_{it} \times PV_{it} + \beta_4 PV_{it}^2 + \beta_5 RivalEntry_{it} \times PV_{it}^2 + store_i + month_t + \varepsilon_{it} \quad (3)$$

5.3 Collocation, Within-Categorical Competition, and Preexisting Competition

Collocation variables ($Collocate_i$), within-categorical competition indicator ($WithinCat_i$), the preexisting competition before entry ($ExComp_i$), and the interaction terms: $RivalEntry_{it} \times Collocate_i$, $RivalEntry_{it} \times PV_{it} \times Collocate_i$, $RivalEntry_{it} \times WithinCat_i$, $RivalEntry_{it} \times PV_{it} \times WithinCat_i$, $RivalEntry_{it} \times ExComp_i$, and $RivalEntry_{it} \times PV_{it} \times ExComp_i$ were added to equation (2), respectively, and formed the following three equations. The coefficient β_5^b indicates the moderating effect of collocation on the effect of entry on sales and is used to test hypothesis 3a. The coefficient β_6^b is for hypothesis 3b and suggests the difference in the effect of product variety adjustment on the entry–sales relationship between collocated and noncollocated stores. Similarly, β_5^c, β_6^c, β_5^d, and β_6^d are estimated to test hypotheses 4a, 4b, 5a, and 5b. Each of these three equations was jointly estimated with equations (1) as a system.

$$\log Sales_{it} = \beta_0^b + \beta_1^b RivalEntry_{it} + \beta_2^b PV_{it} + \beta_3^b RivalEntry_{it} \times PV_{it} + \beta_4^b Collocate_i + \beta_5^b RivalEntry_{it} \times Collocate_i + \beta_6^b RivalEntry_{it} \times PV_{it} \times Collocate_i + Z_{it}B^b + month_t + \varepsilon_{it} \quad (4)$$

$$\log Sales_{it} = \beta_0^c + \beta_1^c RivalEntry_{it} + \beta_2^c PV_{it} + \beta_3^c RivalEntry_{it} \times PV_{it} + \beta_4^c WithinCat_i + \beta_5^c RivalEntry_{it} \times WithinCat_i + \beta_6^c RivalEntry_{it} \times PV_{it}$$

$$\times WithinCat_i + Z_{it}B^c + month_t + \varepsilon_{it} \quad (5)$$

$$\log Sales_{it} = \beta_0^d + \beta_1^d RivalEntry_{it} + \beta_2^d PV_{it} + \beta_3^d RivalEntry_{it} \times PV_{it} + \beta_4^d ExComp_i + \beta_5^d RivalEntry_{it} \times ExComp_i + \beta_6^d RivalEntry_{it} \times PV_{it} \times ExComp_i + Z_{it}B^d + month_t + \varepsilon_{it} \quad (6)$$

Fixed effects for stores were not included after store-varying variables, such as $Collocate_i$, $WithinCat_i$, and $ExComp_i$, were added to the estimation. Otherwise, these variables would not be identified. In addition, to account for store-level variations, some control variables for stores were added in the vector Z_{it}, which includes population ($Population_i$), median household income ($Income_i$).

6 Results

6.1 Product Variety Adjustment in Response to Entry and Its Mitigating Role to Sale Loss

The equation systems were estimated using the three-stage least squares. The estimated results of equations (1) and (2) are reported in Table 3. In the estimated results of equation (1) for product variety, the coefficient for $RivalEntry_{it}$ is significant and positive, which suggests that, on average, incumbent stores increased product variety in response to the entry of warehouse clubs. The instrumental variable $Temperature_{it}$ has a coefficient of 1.959 at the 0.001 significance level, which supports the strength of the instrument. To test the exclusion restriction, the variable was also included in equation (2) and has an estimated coefficient of 0.005 but not significant at the 0.05 level. Hence, the instrument satisfies the exclusion restriction and is excluded from equation (2) in Table 3. Since the fixed effects for year-month and for stores catch most variances over time and across stores, the R^2 values are relatively high in the estimated results for models including fixed effects.

In the estimated results of equation (2) in Table 3, the coefficient for $RivalEntry_{it}$ is significant and negative. It suggests that, on average, sales at the incumbent stores decreased by 17.8% after entry, which supports hypothesis 1. Product variety at stores is positively associated with their sales. The significant and positive coefficient for $RivalEntry_{it} \times PV_{it}$ indicates 0.2% less sales loss at incumbent stores after entry when one unit increased in nonoverlapping product variety. For example, compared to an average 17.8% sales decrease after entry, a store's sales would decrease by 15.8% (calculated by 17.8% – 0.2% × 10) after adding 10 more nonoverlapping SKUs. In relative terms, the decrease in sales after increasing 10 more nonoverlapping SKUs is 88.8% (calculated by 15.8%/17.8%) of the sales decrease at stores with no product variety adjustment. That is, increased product variety reduced sales loss at these incumbent stores after entry. Hypothesis 2a is supported.

Table 3. Value of product variety adjustment on the entry–sales relationship.

Equation	(1)	(2)	(3)
Dependent variable	PV_{it}	$\log Sales_{it}$	$\log Sales_{it}$
$RivalEntry_{it}$	2.037***	−0.178***	−0.276***
	(0.259)	(0.011)	(0.018)
PV_{it}		0.015***	0.026***
		(0.001)	(0.002)
$RivalEntry_{it} \times PV_{it}$		0.002***	0.007***
		(0.000)	(0.001)
PV_{it}^2			−0.00009***
			(9.45×10^{-6})
$RivalEntry_{it} \times PV_{it}^2$			−0.00004***
			(4.50×10^{-6})
$Temperature_{it}$	1.959***		
	(0.043)		
Constant	71.175***	8.370***	8.510***
	(3.121)	(0.136)	(0.121)
Fixed-effects for year-month	Yes	Yes	Yes
Fixed-effects for Store	Yes	Yes	Yes
Observations	17,766	17,766	17,766
R^2	0.958	0.955	0.960

***$p < .001$, **$p < .01$, *$p < .05$.

6.2 Nonlinear Pattern in the Mitigating Role of Product Variety Adjustment to Sale Loss

The estimated results for the nonlinear effects of product variety are also reported in Table 3. The results for equation (1) remain stable. Thus, they are not repeatedly reported for simplification. The results of equation (3) suggest a nonlinear main effect of product variety on sales (positive with a diminishing marginal effect). Besides, the coefficient for $RivalEntry_{it} \times PV_{it}$ is 0.007 and significant, while the coefficient for $RivalEntry_{it} \times PV_{it}^2$ is negative and significant but with a negligible magnitude (−0.00004). This suggests that the mitigation effect of nonoverlapping product variety on sales reduction after the entry is positive with a diminishing return. These results support hypothesis 2b. One explanation for the small effect size of the diminishing return could be that the incumbent stores in the data sample did not increase product variety enough to demonstrate its strong nonlinear moderating effect.

6.3 Collocation, Within-Categorical Competition, and Preexisting Competition

The estimated results of equations (4) to (6) are reported in Table 4. They were estimated in the system with equation (1). Since the estimated results of equation (1) are similar to those in Table 3, they are not reported in Table 4 for simplification. In equation (4), the coefficients for $RivalEntry_{it} \times Collocate_i$ and $RivalEntry_{it} \times PV_{it} \times Collocate_i$ are both significant and positive. These results support hypotheses 3a and 3b. After the

Table 4. Value of product variety adjustment moderated by collocation, within-categorical, and preexisting competition.

Equation	(4)	(5)	(6)
Dependent variable	log $Sales_{it}$	log $Sales_{it}$	log $Sales_{it}$
$RivalEntry_{it}$	−0.181***	−0.486***	−0.198***
	(0.016)	(0.026)	(0.037)
PV_{it}	0.021***	0.024***	0.027***
	(0.000)	(0.000)	(0.000)
$RivalEntry_{it} \times PV_{it}$	0.004***	0.011***	0.003***
	(0.001)	(0.001)	(0.001)
$Collocate_i$	−0.032***		
	(0.008)		
$RivalEntry_{it} \times Collocate_i$	0.025*		
	(0.011)		
$RivalEntry_{it} \times PV_{it} \times Collocate_i$	0.005*		
	(0.002)		
$WithinCat_i$		0.532***	
		(0.015)	
$RivalEntry_{it} \times WithinCat_i$		−0.697***	
		(0.051)	
$RivalEntry_{it} \times PV_{it} \times WithinCat_i$		0.013***	
		(0.001)	
$ExComp_i$			−0.051***
			(0.006)
$RivalEntry_{it} \times ExComp_i$			0.022*
			(0.011)
$RivalEntry_{it} \times PV_{it} \times ExComp_i$			−0.001***
			(0.000)
$Population_i$	0.002***	0.002***	0.004***
	(0.001)	(0.000)	(0.001)
$Income_i$	0.001***	0.001***	0.001***
	(0.000)	(0.000)	(0.000)
Constant	6.633***	6.627***	6.551***
	(0.025)	(0.025)	(0.026)
Fixed-effects for year-month	Yes	Yes	Yes
Observations	17,766	17,766	17,766
R^2	0.730	0.789	0.767

Note. Fixed-effects for stores are not included when store-varying variables are included, such as $Collocate_i$, $WithinCat_i$, and $ExComp_i$. Otherwise, these variables would not be identified.
***$p < .001$, **$p < .01$, *$p < .05$.

entries of warehouse club stores, incumbent stores collocated with the new stores have sales loss less than noncollocated stores due to the agglomeration effect. This finding provides empirical evidence for the agglomeration effect associated with the entrants. Furthermore, increases in nonoverlapping product variety helped the collocated stores alleviate sales loss due to entry more than those at noncollocated stores because increased product variety amplifies the agglomeration effect.

In equation (5), the coefficient for $RivalEntry_{it} \times WithinCat_i$ is significant and negative, but the coefficient for $RivalEntry_{it} \times PV_{it} \times WithinCat_i$ is significant and positive. These results support hypotheses 4a and 4b. After rivals' entries, retail stores in within-categorical competition with new warehouse club stores suffered sales decreases more than incumbent stores in cross-categorical competition with entrants. Furthermore, increases in nonoverlapping product variety helped the within-categorical stores alleviate sales loss due to entry more than those cross-categorical incumbent stores.

Meanwhile, the coefficient for $RivalEntry_{it} \times ExComp_i$ is significant and positive. This suggests that the incumbent stores facing a higher preexisting competition in the market before entry had fewer sales decrease after entry. The coefficient for $RivalEntry_{it} \times PV_{it} \times ExComp_i$ is significant and negative. The preexisting competition softens the mitigation effect of product variety adjustments on sales loss after entry. Thus, hypotheses 5a and 5b are supported.

The results for control variables are as expected in Table 4. Sales at incumbents within the category of the new warehouse club stores are larger than other incumbents. The preexisting competition is negatively associated with sales because competition is negatively related to sales. Population and income are positively associated with sales.

7 Robustness Checks

Several robustness checks are conducted to ensure the findings are not driven by the matching methods, pretreatment time trends, alternative explanations, or market radius selection. First, the estimated results using the dataset with no matching are reported in E-Companion B. Second, time trends and placebo tests are reported in E-Companions C and D. Third, alternative explanations to our findings might be the entry or exit of stores other than the warehouse clubs. These are tested and excluded in E-Companion E. Fourth, the robustness of the findings is tested by reducing the radius of a local market to 5 miles and increasing the radius to 15 miles. In the data sample associated with the newly defined local markets, a 21.6% sales decrease for 5-mile radius and a 7.3% sales decrease for 15-mile radius are found at incumbent stores after entry. Increased product variety mitigates the sales reduction. Although the magnitudes of the effects vary with market radiuses, these results still support the findings.

8 Heterogeneous Effects

8.1 Within-Categorical Stores versus Cross-Categorical Stores

The incumbent retail stores, which are in within- and cross-categorical competition with the entrants, are indicated by a dummy variable in Table 4. To further study how the benefits of product variety adjustment differ between within-categorical and cross-categorical stores, the system of equations (1) and (2) is estimated using two split data samples: one for within-categorical stores and the other for cross-categorical stores. The estimated results are summarized in Table 5. Product variety at incumbent within-categorical stores increased, while

Table 5. Heterogeneous effects of product variety adjustment across retail categories.

	Within-categorical stores		Cross-categorical stores	
Equation	(1)	(2)	(1)	(2)
Dependent variable	PV_{it}	$\log Sales_{it}$	PV_{it}	$\log Sales_{it}$
$RivalEntry_{it}$	4.858***	−0.356***	0.053	−0.089*
	(1.923)	(0.079)	(0.046)	(0.037)
PV_{it}		0.094***		0.062***
		(0.028)		(0.008)
$RivalEntry_{it} \times PV_{it}$		0.015***		0.012
		(0.004)		(0.012)
Observations	4,986		12,780	

Note. Fixed-effects and other variables are included in all estimations.
***$p < .001$, **$p < .01$, *$p < .05$.

product variety at cross-categorical stores did not change significantly. The sales at all incumbent stores decreased after entry. Increases in nonoverlapping product variety mitigate sales decreases at within-categorical stores. At cross-categorical stores, increased nonoverlapping product variety does not reduce the sales loss due to entry. These findings support the theory of category and competition, where firms in different categories are suggested to take different strategies to compete (Cattani et al., 2017). Following a rival's entry, adjusting product variety does not necessarily help mitigate the sales decreases for all incumbents.

8.2 *Entries of Different Warehouse Club Chains*

In the time window and geographic locations included in the data sample, two warehouse club chain entries were identified: Costco and BJ's. Due to the different market shares and member bases of these two chains, the impacts of their entries on incumbents' sales may be different. A dummy variable of $CostcoEntry_i$ was generated to indicate the incumbent stores whose local markets include a new Costco store. The base case is for incumbent stores whose local markets include new BJ's stores. Then, the variables $RivalEntry_{it} \times CostcoEntry_i$ and $RivalEntry_{it} \times PV_{it} \times CostcoEntry_i$ were added to equation (2). The estimated results are reported in Table 6.

The incumbents that have a new Costco store in the local markets, on average, have more sales than those with a new BJ's store in the markets. However, the negative impact of a Costco's entry on sales at incumbents is larger than a BJ's entry. Furthermore, increasing product variety after a Costco's entry mitigates the sales reduction at incumbent stores more than that after a BJ's entry. One explanation is that a BJ's store (around 7,000 SKUs) carries a greater variety of products than a Costco store (around 4,000 SKUs) (Perry, 2022). Increases in incumbents' product variety create fewer nonoverlapping products with BJ's new stores than with Costco's new stores.

Table 6. Heterogeneous effects for Costco's versus BJ's entries.

Dependent variable	PV_{it}	$\log Sales_{it}$
$RivalEntry_{it}$	1.150***	0.133***
	(0.274)	(0.031)
PV_{it}		0.028***
		(0.001)
$RivalEntry_{it} \times PV_{it}$		0.003**
		(0.001)
$CostcoEntry_i$		0.084***
		(0.014)
$RivalEntry_{it} \times CostcoEntry_i$		−0.303***
		(0.038)
$RivalEntry_{it} \times PV_{it} \times CostcoEntry_i$		0.004***
		(0.001)
Fixed-effects for year-month	Yes	Yes
Observations	17,766	17,766
R^2	0.952	0.764

Note. Fixed-effects for stores are not included in the equation for sales. Otherwise, $CostcoEntry_i$ would not be identified.
***$p < .001$, **$p < .01$, *$p < .05$.

Thus, increased product variety at incumbent stores has a mitigation effect on the influence of a new BJ's store less than a new Costco store.

9 Managerial Implications

The results provide numerous managerial implications for managers of retail firms facing intensified competition caused by the market entry of major competitors. The results illustrate the substitution and agglomeration effects of warehouse club's entries by studying the incumbent store locations. Due to the trade-off between the substitution and agglomeration effects, incumbent stores collocated with new warehouse club stores suffer a reduction in sales after the rival's entry, which is less than incumbent noncollocated stores.

In general, increased product variety at incumbent stores after entry generates sales value by weakening the negative impact of entrants on sales at incumbents. However, simply increasing product variety after rivals' entries does not guarantee sales benefits. For example, increased product variety at incumbent stores in cross-categorical competition with entrants did not mitigate the sales loss due to a rival's entry. Moreover, the sales benefits of product variety adjustments after entry diminish with increases in the preexisting competition in the market. In a less competitive market, increasing product variety after entry can be an effective decision for incumbents. Nevertheless, increased product variety after rivals' entries may not work well in a highly competitive market where consumers are not sensitive to additional products. Thus, incumbent stores should not blindly follow a high product variety decision after a rival's entry. Instead, incumbents are suggested to consider whether the entrants are rivals and the preexisting competition.

In addition, the main effects of rivals' entries on incumbents' sales and the mitigation effects of product variety adjustments vary across different warehouse club chains. For example, Costco and BJ's entries result in different impacts on sales at incumbents. Further, the incumbents' product variety adjustments generate heterogeneous benefits across the entries of these two chains. Even when facing competition from an entrant in the same category, incumbents should consider the entrant's features to decide whether and by how much to adjust product variety.

10 Conclusion, Contributions, and Limitations

This paper analyzed the impact of incumbent retail stores' product variety adjustments on their sales loss after warehouse clubs' entry. Through the analyses of incumbent stores collocated with new warehouse club stores, this empirical study contributes to the literature on market entry and competition by disentangling the substitution and agglomeration effects associated with a market entry. While substitution and the agglomeration effects have been used to explain the influences of location on firms' response to competition (McCann and Vroom, 2010; Ren et al., 2011), limited empirical evidence for these effects has been provided. The results provide empirical evidence for the substitution and agglomeration effects associated with the entry of warehouse clubs.

The literature suggests that incumbents respond to market entry by adjusting product variety (Ren et al., 2011, 2019; Seamans and Zhu, 2014; Watson, 2009), and product variety decisions are found to affect sales in stable markets (Brynjolfsson et al., 2003; Kekre and Srinivasan, 1990; Lancaster, 1990). However, the lack of empirical studies on the sales benefits of product variety adjustments after rivals' entries limits the understanding of decisions after changes in competition. The sales affected by product variety adjustments after rivals' entries are not straightforward because both the change in consumer demand due to rivals' entries and the incumbent retailers' product variety decisions simultaneously affect incumbents' sales. This study contributes to the literature by exploring the sales benefit of the incumbent's product variety decisions to rivals' entries. Although incumbents often vary their product variety in response to rivals' entries, increasing product variety following the entry does not guarantee sales benefits to all incumbents. For example, convenience and drug stores did not benefit from increased product variety by reducing the sales loss to new entrants. An incumbent's decision on product variety adjustments following a rival's entry needs to consider its competition category relative to the entrant and preexisting competition in the market.

This paper also contributes to the literature by exploring the nonlinearities in the impacts of product variety. While the literature identifies the direct impact of product variety on sales (Brynjolfsson et al., 2003; Ramdas, 2003; Wan et al., 2012), this study extends the literature by uncovering the nonlinear pattern of the moderating role of product variety in the entry–sales relationship. Although the diminishing return of product variety is negligible in this data sample, the results identify that increasing nonoverlapping product variety reduces the sales loss after rivals' entries with a diminishing return. This nonlinearity could be stronger in the context of greater product variety adjustments.

Further, the empirical analyses in this paper contribute to the literature on categories and competition (Cattani et al., 2017; Suarez et al., 2015). Scholars studied categorical competition at the industry level, such as the automobile industry (Nohria and Garcia-Pont 1991). This study extends the literature by studying the effects of product variety adjustments in the context of within- and cross-categorical competition at the firm/retail store level. The similarities and differences between firms affect their decisions and the outcomes of these decisions. This paper sheds light on the categorical competition at the firm level by studying how incumbent retail stores' product variety adjustments reduce sales loss after rivals' entries. The benefits of product variety adjustment vary from within-categorical competition to cross-categorical competition.

This study is limited by the data availability. In the study's empirical context, the incumbents increased product variety to raise nonoverlapping products with new warehouse club stores that carry a limited variety of popular products. Future research could benefit from examining the impact of other types of entrants, such as supermarkets and department stores. Additionally, the results might be specific to the product categories included in the analysis. Effects and trade-offs associated with product variety decisions may be different across product categories. The empirical context of this study does not provide an opportunity to study this. Moreover, other exciting future research topics include the long-term effects of entries, the customer bases of incumbent retail stores, profitability, pricing, and the recent digital transformation in the retailing industry. Although the effect size in this study may vary when the empirical context varies (like most empirical findings in the literature), this paper provides empirical evidence for the existence and the patterns of the mitigation role of product variety in the rival entry–incumbent sales relationship. These effects and patterns provide insights when scholars study related topics in other empirical contexts.

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